MIL-STD-1465A(AR)
4 AUGUST 1987

SUPERSEDING
MIL-STD-1465A(AR)
30 SEPTEMBER 1982

MILITARY STANDARD

CONFIGURATION MANAGEMENT OF ARMAMENTS, MUNITIONS, AND CHEMICAL PRODUCTION BASE MODERNIZATION



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DEPARTMENT OF DEFENSE
Washington, DC 20301

CONFIGURATION MANAGEMENT OF ARMAMENTS, MUNITIONS, AND CHEMICAL PRODUCTION BASE MODERNIZATION

MIL-STD-1465A(AR)

1. This military standard is approved for use by the US Army Production Base Modernization Activity (PBMA), Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, US Army PBMA, ATTN: AMSMC-PBT-C, Dover, NJ 07801-5001, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FOREWARD

1. This Military Standard, was developed to apply Configuration Management to the Army Armaments Munitions, and Chemical Production Base Modernization Program. It is an expansion of the former MIL-STD-1465(AR) which covered only the Configuration Management of the Modernization of the Munitions Production Base. It contains the policies, responsibilities, and operating procedures for implementing the program.

2. The intent of this standard is to provide a uniform and tailored configuration management application to a program that is not systems/materiel/item oriented, but is primarily based upon facility construction and process equipment.

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1: SCOPE

1.1 Purpose. This standard establishes the policy, responsibilities, and procedures for implementing configuration management to facilities and equipment in support of the Munitions, Armaments and Chemical Production Base Modernization (PBM) program. Software is not included in this MIL-STD and will be addressed under separate cover.

1.2 Application. The provisions of this standard apply to all participating Government agencies and will be imposed, as applicable, on participating contractors. This standard applies through the life cycle of all modernization projects. Any exceptions shall be directed or approved by the Commander PBMA.

2: REFERENCED DOCUMENTS

2.1 Government Documents

2.1.1 Specifications, Standards, and Handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this standard to the extent specified herein.

Standards
Military
MIL-STD-490 - Specification Practices

Handbooks
Cataloging Handbook H4 - Manufactures Federal Item Identification Guides for Supply Cataloguing

2.1.2 Other Government Document, Drawings, and Publications. The following other Government documents, drawings, and publications form a part of this standard to the extent specified herein.

TM5-800-3 - Project Development Brochures Part I and Part II, Functional Requirements

3: DEFINITIONS

3.1 Baseline. A reference point to which all changes must be specifically related.

3.2 Concept Baseline. The concept baseline consists of the approved facility concept design. The Concept Baseline must be established prior to initiation of final design.

3.3 Configuration. The functional and physical characteristics of equipment, facilities, and process control systems as defined in technical documentation.

3.4 Configuration Accounting. The reporting and recording of configuration baselines and changes thereto, together with associated change implementation and impact status.

3.5 Configuration Baselines. Configuration reference points established by approval of technical documentation. All changes must be specifically related to these reference points. Configuration baselines for facilities consist of design criteria, concept design, and final design. Configuration baselines for equipment consist of process design, functional criteria, and technical data package (TDP). The documentation and data that comprise the operational baseline will apply to both facilities and equipment as required.

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3.6 Configuration Control. The systematic evaluation, coordination, and approval or disapproval of all proposed configuration changes and verifications of approved change incorporation after the formal establishment of baselines.

3.7 Configuration Control Board (CCB). A functional committee, composed of representatives from organizations affected by changes, which reviews and approves or disapproves proposed changes within its authority.

3.8 Configuration Identification. The documentation defining the approved configuration of equipment, or a facility.

3.9 Configuration Item (CI). Hardware, or any of its discrete portions, which satisfies an end use function and is designated by the Government for configuration management. CIs may vary widely in complexity, size, and type, from a facility or production line to an individual piece of process equipment.

3.10 Configuration Management. A discipline to properly identify, control changes to, record, and audit the change implementation status of the total configuration of equipment and facilities.

3.11 Criteria Baseline. The configuration baseline consists of the approved facility design criteria. The Criteria Baseline must be established prior to initiation of design. The facility design criteria is synonymous with the Project Development Brochure as outlined in TM5-800-3.

3.12 Design Baseline. The configuration baseline established upon award of construction contracts and consisting of interfaces and related features of the final design.

3.13 Engineering Change. Any approved change in design or performance (forms, fit or function) of an item, system, or facility delivered. Such changes will require change or revision to the specification, engineering drawing, or the documents referenced therein.

3.14 Engineering Change Proposal (ECP). An ECP is used to propose, identify, and establish a partial or complete baseline or to effect engineering changes.

3.15 Executive Secretary. An individual assigned the responsibility for administrative functions and technical coordination to support the CCB.

3.16 Functional Baseline. The configuration baseline consisting of the functional criteria specification for equipment.

3.17 Operational Baseline. The configuration baseline established at the culmination of a PBMA facility project. It is a list of applicable documentation that identifies the project plus operations and maintenance (O&M) data.

3.18 Process Baseline. The configuration baseline established upon the acceptance of a pilot line or equipment prototype, consisting of the technical data package (TDP) developed for the pilot line, prototype, and follow-on facility. Usually the result of an MMTE project.

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3.19 Routine ECP. An ECP which will be processed through normal channels and with a normal review period.

3.20 Specification. A document which clearly and accurately describes the essential technical requirements for items, materials, or services including the procedures by which it will be determined that the requirements have been met.

3.21 Technical Baseline. The configuration baseline consisting of the equipment TDP.

3.22 Technical Data Package (TDP). A technical description of an item adequate for use in procurement and subsequent operation and maintenance. The description defines the required design configuration and assures adequacy of performance. It consists of all applicable technical data.

3.23 Urgent ECP. An ECP requiring expeditious handling and approval to avoid schedule slippage or increased costs.

3.24 Value Engineering (VE). A systematic effort directed at analyzing the functional requirements of systems, equipment, facilities, procedures, and supplies for the purpose of achieving the essential functions at the lowest total cost, consistent with the needed performance, reliability, safety, quality, and maintainability.

3.25 Value Engineering Change Proposal (VECP). The ECP submitted by a non-government proposer which, if approved, will result in a savings to the Government. In the case of Corps of Engineers (CE) construction or supply contracts, the VE proposal is made by letter and the CE prepares the VECP for submission to the CCB.

3.26 Value Engineering Proposal (VEP). The ECP submitted by a Government proposer which, if approved, will result in a savings to the Government.

3.27 Vendor Data. Information such as description, installation, operation and maintenance, data furnished by the manufacturer of an item of equipment.

3.28 Definition of Acronyms/Abbreviations Used In This Standard. The following acronyms listed in this Military Standard are defined as follows:

a. AAP Army Ammunition Plant

b. AMC Army Materiel Command

c. AMCCOM US Army Armament Munitions and Chemical Command

d. AR Army Regulation

e. ARDEC US Army Armament Research, Development, and Engineering Center

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f.	B/L	Baseline
g.	BOD	Beneficial Occupancy Date
h.	CCB	Configuration Control Board
i.	CDRL	Contract Data Requirements List (DD Form 1423)
j.	CE	Corps of Engineers
k.	CI	Configuration Item
l.	COCO	Contractor Owned-Contractor Operated
m.	COR	Commanding Officers Representative
n.	CWE	Current Working Estimate
o.	DA	Department of the Army
p.	DL	Data List
q.	DOD	Department of Defense
r.	ECP	Engineering Change Proposal
s.	ER	Engineer Regulation
t.	EXP	Expansion Project
u.	FCCB	Field Configuration Control Board
v.	GBL	Generation Breakdown List
w.	GOCO	Government-Owned - Contractor-Operated
x.	GOCO	Government-Owned - Government-Operated
y.	IBEA	Industrial Base Engineering Activity
z.	IFB	Invitation for Bid
a.a.	IPIPMO	Industrial Preparedness and Installation Program Management Office
b.b.	I&SA	US Army Materiel Command, Installations & Services Activity, Rock Island
c.c.	IR	Industrial Readiness Directorate, Rock Island
d.d.	LAP	Load, Assemble, and Pack

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e.e.	MIL-STD	Military Standard
f.f.	MOD	Modernization Project
g.g.	MMTE	Manufacturing Methods and Technology Engineering
h.h.	MPTS	Metal Parts
i.i.	NA	Not Applicable
j.j.	OCE	Office, Chief of Engineers
k.k.	O&M	Operations and Maintenance
l.l.	OSD	Office, Secretary of Defense
m.m.	OSHA	Occupational Safety and Health Act
n.n.	PAL	PS&ER, Armament and Layaway
o.o.	P&E	Propellants and Explosives
p.p.	PBM	Production Base Modernization
q.q.	PBMA	Production Base Modernization Activity
r.r.	PBS	Production Base Support
s.s.	PCO	Procuring Contracting Officer
t.t.	PEP	Plant Equipment Package
u.u.	PL	Parts List
v.v.	RAM	Reliability, Availability, and Maintainability
w.w.	ROM	Rough Order of Magnitude
x.x.	SCN	Specification Change Notice (DD Form 1696)
y.y.	SOW	Scope of Work
z.z.	TDP	Technical Data Package
a.a.a.	TDPL	Technical Data Package Listing
b.b.b.	TM	Technical Manual
c.c.c.	USAEDH	US Army Engineer Division, Huntsville

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d.d.d. VE Value Engineering

e.e.e. VECP Value Engineering Change Proposal (Contractor submitted)

f.f.f. VEP Value Engineering Proposal (in-house)

4: GENERAL REQUIREMENTS

4.1 Objectives. The objectives of this standard are to allow optimum latitude in criteria development and design, yet introduce at the appropriate time the degree and depth of identification, configuration control, status accounting and audit to support the Configuration Management of the Army's Production Base Modernization.

4.2 General Policy.

4.2.1 Relationship Between Criteria Developer and Designer. The objectives outlined above are achieved through a close working relationship between the criteria developer and the designer during the entire criteria, design, and execution cycle. This standard provides the framework for such a relationship while preserving the basic responsibilities of each participating agency. The standard has been prepared to be readily understood and utilized by personnel without previous experience in formal configuration management.

4.2.2 Configuration Management Procedures. This standard establishes the configuration management procedures among agencies and does not include procedures within each agency. It is anticipated that each agency will prepare its own internal configuration management plan and will designate a senior individual responsible to his commander for configuration management within the agency and to serve as a contact point between agencies.

4.2.3 Configuration Management Concepts. Standard DOD Configuration Management is primarily systems/materiel/item oriented with very little direct applicability to facility construction or process equipment development for the production base. The basic concepts of identification, control, accounting, and audit can be applied, however, to facility construction as well as to process equipment development. The nomenclature is different, but the basic goals are the same. Inasmuch as this standard is to be utilized primarily by personnel who are not familiar with formal configuration management procedures and nomenclature, the procedures and nomenclature have been simplified insofar as practical. The forms and reports can be readily understood and utilized with a minimum of reference to the instructions. This will permit a broad participation in the program with a minimum need for training and indoctrination.

4.2.4 Implementation. The PBMA configuration management program is based on the concept of establishing approved baselines, controlling changes to these baselines, recording and auditing the approved configuration. Directives and procedures provide for:

a. Early establishment of configuration baselines.

b. Expeditious processing of proposed changes to include coordination with all affected agencies.

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c. Expeditious decision by the CCB or appropriate management level where necessary.

d. Timely incorporation of approved changes.

e. Timely visibility of change status to management at all levels.

f. Timely revision of affected documentation to reflect current configuration.

4.2.5 Changes. Changes to the baseline configuration, once established, will be minimized and will generally be made only in the interest of:

a. Correction of deficiencies or resolution of conflicts as required to achieve operational requirements.

b. Significant improvement in operation or performance.

c. Incorporation of advances in the state-of-the-art.

d. Compliance with revisions in environmental standards (OSHA).

e. Safety.

f. VE.

g. Cost Reduction/Cost Avoidance.

5: DETAILED REQUIREMENTS

5.1 Configuration Identification.

5.1.1 General

5.1.1.1 Project Development. In order to determine the necessary reference points for configuration identification during the life cycle of a project, it is necessary to understand the overall development phases of a standard project. Figure 1 depicts the development for a typical project which contains both equipment and facilities construction work. Each of these phases will be further defined and amplified in this section.

5.1.1.2 Project Baselines. Configuration identification will be established by formal project baselines representing significant points in project development. These baselines are identified below as to specific equipment and facilities application. A sequential flow of project baselines is outlined in Figure 2.

5.1.1.3 Facility Identification. Configuration identification for facilities will consist of the facility design criteria documentation for each project, concept design, final design, and the operational data to be incorporated into the operational data package. The funding program documents will not be controlled by formal configuration management. The early preparation of comprehensive design criteria documentation to identify the requirements of the using agency is one of the primary objectives of this MIL-STD. The CE will assist and participate in design criteria preparation to the extent authorized by the PBMA.

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5.1.1.4 Equipment Identification. Configuration identification for equipment will consist of the equipment pilot line or prototype documentation for each project, as applicable, the equipment functional criteria, the equipment TDP, and the operational data to be incorporated into the operational data package. The early preparation of complete and comprehensive functional criteria documentation is essential for defining performance requirements, describing processes, and establishing design requirements of the using agency. It is highly desirable that the design of process equipment be accomplished well in advance of the development of facilities design criteria; however, it is usually only possible in those cases where the equipment has been developed and proven under an MMTE program and drawings and specifications suitable for procurement are available. In those cases where the suitable drawings and specifications are not available, the design effort for development of functional criteria must be limited to a definition of characteristics, e.g., equipment dimensions, RAM characteristics, nominal weight, clear aisle and overhead, type and capacity of utilities available or to be furnished, etc. This definition of characteristics is necessary to permit development of facility design criteria, at the least possible cost, by avoiding under or over-design of building and utilities.

5.1.2 Procedures for Facility Identification.

5.1.2.1 Criteria Baseline. The Project Development Brochure, Constituting the Criteria Baseline, will normally be prepared in response to direction from the PBMA. The procedures and format will follow the guidance contained in TM 5-800-3. Interface requirements of the using agency such as clear height, aisle width, utility services required, and mounting details for equipment to be installed will be clearly identified. For utility services to equipment, the locations of the interface points are to be identified as well as the character and quantity of the services. Interfaces not identified in the design criteria document but to be furnished with the equipment vendor data shall be stated. Because the Project Development Brochure contains the actual requirements of the using agency, it will be controlled and maintained until project completion. The Project Development Brochure will be the point of reference in the evaluation of criteria change ECP when established requirements are affected.(see 6.2)

5.1.2.2 Concept Baseline. The concept design for facilities will be prepared by the CE upon receipt of a directive from OCE. The procedures and format will follow normal CE practice for preparation of concept documents. Concept design will be prepared in sufficient detail to permit adequate review by the using agency to verify the design approach and to permit the preparation of a firm cost estimate by the CE to be utilized by AMCCOM for programming the cost of the project. The concept baseline is superseded upon approval of the design baseline. Data for the Concept Baseline will be obtained by CE utilizing their procedures. CE data acquisition procedures are not addressed in this MIL-STD.

5.1.2.3 Design Baseline. The final design for facilities will be prepared by the CE upon receipt of a directive from OCE. The procedures and format will follow normal CE practice. The design baseline is automatically established upon award of the construction contract. For configuration management record purposes, a baseline ECP will be prepared by USAEDH indicating that the design has been baselined as of the last amendment to the construction bid documents. Participating agencies who maintain a set of the design documentation will be on the distribution list for the IFB and amendments

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as they are issued to the bidders; therefore, the design documentation will not be reproduced and distributed again with the baseline ECP. Data for the Design Baseline will be obtained by CE utilizing their procedures. CE data acquisition procedures are not addressed in this MIL-STD.

5.1.2.4 Exception to Standard Facility Identification. Separate instructions will be individually issued by PBMA for projects which, because of contract applications, are not covered by standard identification provisions as outlined in this standard. These would include "two-step", and "cost-plus-fixed-fee" type contracts administrated by the CE.

5.1.3 Procedures for Process Equipment Identification.

5.1.3.1 Process Baseline. The TDP for a pilot line or equipment prototype normally will be developed by ARDEC or another engineering activity as a result of an MMTE project. The process baseline will be established only for those projects for which a pilot line or equipment prototype has been developed. The data for a process baseline is obtained by ARDC or another engineering activity as part of the MMTE effort and is not a data requirement to be obtained under provisions of this standard.

5.1.3.2 Functional Baseline. The equipmnent functional criteria specification, constituting the Functional Baseline, will be prepared in response to direction from the PBMA. When the project also involves a facility effort, the functional baseline will be established as early as possible, but not later than the approval of the criteria baseline. The equipment functional criteria specification for a project will consist of a functional layout of the process system describing the physical and functional interface, a narrative description of the processes, a list of equipment CIs, and a complete list of references to descriptive technical documentation. The CIs will be identified as separate entities, each physically separable from the system and capable of being procured and installed as a separate CI. The equipment functional criteria specification preparation will be based on knowledge of production requirements and of present and newly available process machinery designs. The equipment functional criteria specification will define the necessary design and performance of the process equipment. Preparation of the equipment functional criteria specification is covered in Appendix A. The PBMA will prepare a functional baseline ECP for submittal to the PBMA CCB for approval when criteria are complete. The equipment functional criteria specification will be maintained in a current status by ECP action until procurement and acceptance of the equipment have been completed. (see 6.2)

5.1.3.3 Technical Baseline. The equipment TDP specification, constituting the Technical Baseline, will be prepared in response to direction from the PBMA. The Technical Baseline will be comprised of a complete equipment TDP (see definition for TDP). Preparation of the equipment TDP specification for individual CIs is covered in Appendix B. The complete TDP specification for a project will consist of a cover sheet as shown in Figure 8; a TDP Listing (TDPL), to include title, specification number, and date for each CI; and the individual TDP specifications for each listed CI. In exception cases, a non specification type TDP may be prepared in contractor or other acceptable format to comprise the Technical Baseline. The PBMA will prepare a Technical Baseline ECP for submittal to the PBMA CCB. The Technical Baseline will be established as a prerequisite

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to preparation of the equipment procurement documentation. When the project also involves a facility effort, the Technical Baseline will be established prior to the CE 90 Percent Design Review Conference. The Technical Baseline will be maintained in a current status by ECP action until project completion. (see 6.2)

5.1.4 Procedures for Project Documentation.

5.1.4.1 Operational Baseline. The Operational Baseline listing, constituting the operational baseline, identifies the documentation procured/developed during project execution that provides the user with the information necessary for identification and effective O&M capability. Typical examples of data items that may be listed are provided below for information only: (see 6.2)

a. "As built" facility drawings.

b. Equipment drawings (production and facility).

c. TDP.

d. Specifications.

e. Installation drawings.

f. Line layout drawings.

g. Manuals (operational, maintenance, training, layaway, reactivation, trouble shooting, supply, safety).

h. Equipment operating procedures.

i. Overall operating procedures.

j. Staffing guide.

k. Skills guide.

l. Test procedures.

m. Tooling lists.

n. Spare/repair parts lists.

o. Provisioning guides.

p. Description of manufacture (DOM).

q. Critical process description.

r. Master equipment list.

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5.2 Configuration Control

5.2.1 General.

5.2.1.1 Configuration Control Boards. Configuration control of established baselines will be exercised through formally established CCBs. The PBMA CCB will control changes to projects for both facilities and equipment. Field level CCBs (FCCBs) will be established at designated locations by direction of the PBMA Commander only.

5.2.1.2 Pre-CCB Coordination. Pre-CCB coordination of ECPs is encouraged, particularly where the impact is major or the proposed solution to the problem is in doubt. The use of preliminary ECPs for coordination is encouraged where preparation of the formal ECP is being delayed for some valid reason. Urgent ECPs with less than normal review periods may also be utilized where the reduction in processing time is required to minimize schedule impact. If necessary and with the consent of the CCB Chairman, urgent ECPs may be approved by the CCB prior to preparation of the formal impact sheet. The objective in all cases is to allow adequate time to review the ECP and to obtain the CCB decision in a timely manner as required to minimize schedule impact. Pre-CCB coordination of simple ECPs is not mandatory and in many cases would only delay the final CCB decision. Lower level working groups will not be established on a permanent basis for pre-CCB coordination. Where possible, objections or planned nonconcurrences by CCB members shall be resolved prior to the CCB meeting. A reasonable amount of across-the-table discussion at the CCB meeting is encouraged; however, the CCB member representing each agency must be prepared to discuss each ECP and must have sufficient authority to commit his agency in such discussions. Any agency planning to nonconcur in an ECP shall advise the originator of the ECP and the Executive Secretary of the CCB at least one day prior to the meeting. All nonconcurrences by members at a CCB meeting must be supported in writing to assist in evaluation and decision by the PBMA Commander in the event of a split board.

5.2.1.3 ECP Preparation and Submission. ECPs may be prepared by any participating agency desiring a change to baselined documentation. ECPs will not be prepared to document no-cost editorial type corrections. ECPs initiated by commercial contractors should be submitted through the PCO to the Executive Secretary of the PBMA CCB. GOCO contractors shall prepare ECPs for submission and signature by the COR staff. The desired change is to be shown in general form allowing final design latitude. If the ECP is approved, the final design to implement it is the responsibility of the design agency which will design directly against the approved ECP, without the necessity of preparing another ECP as authority to revise the design. (see 6.2)

5.2.1.4 Classified Enclosures. Classified information will not be included in an ECP. Classified enclosures will be transmitted separately through normal security channels and referenced in the ECP.

5.2.2 PBMA CCB.

5.2.2.1 CCB Board Members. The PBMA CCB, will be located at PBMA, Dover, NJ, which will provide the Executive Secretary to perform the staff work associated with the CCB and a nonvoting Chairman to conduct the meetings. The PBMA CCB shall be comprised of members from the following agencies:

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a. PBMA

b. ARDEC

c. IPIPMO

d. IBEA

e. I&SA

f. USAEDH

g. IR

h. Other participating agencies identified by the PBMA

The above list is general in scope and this standard will not be revised to maintain a specific list. Ad hoc members may be appointed by the Chairman to review any ECP and each will have a vote concerning those ECPs for which they have been designated to participate at the meeting. Technical advisors may accompany any board member and may be called upon for advice and information, but will not have a vote concerning any ECP.

5.2.2.2 Delegation of Authority. The Chairman, acting with the authority of the PBMA Commander, may approve individual ECPs with estimated dollar impact not to exceed an increase of $200,000, provided that all members are in agreement and that this dollar impact does not exceed the available funding for the individual project during the execution phase. ECPs will not be divided to circumvent ceiling restructions and, the Chairman shall consolidate related ECPs, if appropriate, to insure that the CCB's authority is not exceeded. No dollar limitation will be placed upon changes to be made prior to the execution phase nor to cost decreases during any phase of the project.

5.2.2.3 Resolution. ECPs will be referred to the PBMA Commander for resolution when:

a. The estimated dollar impact exceeds the PBMA CCB authority.

b. The vote results in a split board. The board is considered to be split when a unanimous decision is not reached.

5.2.3 Other CCBs. FCCBs or special CCBs will be the only subordinate CCBs established. An FCCB or special CCB may be established at the discretion of the PBMA Commander. A charter will be issued to each CCB when it is established.

5.2.4 ECP Preparation.

5.2.4.1 Examples. DA Form 5198-R (Production Base Engineering Change Proposal) (ECP) will be utilized as the ECP form for both baselining and controlling changes to equipment and facilities. DA Form 5198-R, DA Form 5199-R (Production Base Modernization Program Engineering Change Proposal (ECP) Continuation Sheet), DA Form 5201-R (Configuration Control Board Engineering Change Proposal (ECP) Concurrence Sheet) may all be locally reproduced on 8½ by 11 inch paper.(see 6.2)

5.2.4.2 ECP Numbering System.

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FORM APPROVED
OMB NUMBER 0704–0188
NO EXPIRATION DATE

(SEE MIL-STD-1465A (AR) FOR INSTRUCTIONS)

PRODUCTION BASE ENGINEERING CHANGE PROPOSAL (ECP)

SHEET 1 OF

1.	a. ECP NUMBER DATE	b. REVISION DATE	c. CORRECTION DATE	d. TYPE ☐ CHANGE ☐ BASELINE	e. PRIORITY ☐ ROUTINE ☐ URGENT	f. APPROVAL NEED DATE (da/mo/yr)	g. CATEGORY ☐ (A) STATE OF THE ART ☐ (B) BASELINE ☐ (D) DESIGN IMPROVEMENT ☐ (O) OPERATIONAL REQUIREMENT ☐ (P) POLLUTION ABATEMENT ☐ (R) COST REDUCTION ☐ (S) SAFETY ☐ (V) VALUE ENGINEERING

2.	a. SITES (name & state)	b. PROJECT NO AND TITLE	c. SP	d. COMMODITY ☐ LOAD, ASSEMBLE AND PACK ☐ METAL PARTS ☐ PROPELLANTS AND EXPLOSIVES ☐ PAL ☐ OTHER (SPECIFY)

3.	ECP TITLE (not to exceed 40 spaces)

4.	DESCRIPTION AND JUSTIFICATION

5.	PRIMARY DOCUMENTATION AFFECTED:		
	a. DOCUMENT NUMBER AND TITLE	b. CURRENT REVISION	c. SHEET/PAGE NUMBERS

6.	RELATED ECP

7.	a. ORIGINATOR/DESIGN ENGINEER	b. SUBMITTED BY:
	NAME _____	SIGNATURE _____
	ORGANIZATION _____	ORGANIZATION _____
	PHONE _____	DATE _____

DA FORM 5198-R, JUL 86

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FORM APPROVED
OMB NUMBER 0704–0188
NO EXPIRATION DATE

(SEE MIL-STD-1465A (AR) FOR INSTRUCTIONS)

PRODUCTION BASE ENGINEERING CHANGE PROPOSAL (ECP)
CONTINUATION SHEET

ECP NUMBER	PROJECT NO.	REVISION	CORRECTION	SHEET NO. ___ OF ___

DA FORM 5199-R, JUL 86

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FORM APPROVED
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(SEE MIL-STD-1465A (AR) FOR INSTRUCTIONS)

CONFIGURATION CONTROL BOARD
ENGINEERING CHANGE PROPOSAL (ECP) CONCURRENCE SHEET

		ECP AS SUBMITTED TO CCB		ECP AFTER CCB ACTION	
1. a. ECP NUMBER DATE	b. PROJECT NUMBER	c. REVISION DATE	d. CORRECTION DATE	e. REVISION DATE	f. CORRECTION DATE
2. CONFIGURATION CONTROL BOARD MEETING			a. CCB PBM	b. MEETING NO.	c. DATE
3. ESTIMATED IMPACT SUMMARY		a. FACILITY	b. EQUIPMENT	c. TOTAL	d. KEY TIME ☐ YES ☐ NO

4. MODIFICATION(S) TO SUBMITTED ECP ☐ NONE ☐ AS FOLLOWS

5. BOARD MEMBERS POSITION ON ECP AS MODIFIED ABOVE

ORG	SIGNATURE	CONCUR YES	CONCUR NO	ORG	SIGNATURE	CONCUR YES	CONCUR NO
AMXEN-IC				BP(D)			
USAEDH							
AMXIB-PE				PB			
ARDC							

6. ACTION TAKEN ON ECP

☐ APPROVED ☐ DISAPPROVED ☐ OTHER (SEE REMARKS)	☐ COMPLETED ACTION ☐ CLEAN-UP ECP REQUIRED ☐ FURNISH IMPACT	SIGNATURE OF CHAIRMAN	DATE

7. REMARKS

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5.2.4.2.1 ECP Designators for Government-owned Plants. ECP numbers are assigned by the agency originating the ECP. The ECP numbering system is designed to indicate the site affected and the agency initiating the ECP. Each ECP number is a seven character alphanumeric group, for example, FKM0006. The first two characters indicate the site affected. In the example above, the site is Radford. These designators are listed below for convenient reference:

ALABAMA	GE	IOWA	F9	PINE BLUFF ARSENAL	FJ
BADGER	GF	JOLIET	FA	RADFORD	FK
BURLINGTON	GK	KANSAS	FY	RAVENNA	G2
CORNHUSKER	FX	LAKE CITY	FB	RIVERBANK	GC
CRANE	M3	LONE STAR	FC	ROCK ISLAND ARSENAL	M5
GATEWAY	EB	LONGHORN	FD	ROCKY MOUNTAIN ARSENAL	FL
HAWTHORNE	DJ	LOUISIANA	FE	SCRANTON	FM
HAYS	TN	McALESTER	DH	ST LOUIS	G4
HOLSTON	F7	MILAN	FF	SUNFLOWER	G3
INDIANA	F8	MISSISSIPPI	MS	TWIN CITIES	FN
INDIAN HEAD	RR	NEWPORT	FG	VOLUNTEER	G5
				WATERVLIET	M7

5.2.4.2.2 Commercial Sites. The designator CX is used in this standard where an ECP applies to a commercial site.

5.2.4.2.3 Undetermined Sites. The designator XX is used in this standard where an ECP applies to an undetermined site. The site may be commercial or Government-owned, but may not be determined prior to design completion.

5.2.4.2.4 Originating Agency. The third character indicates the agency originating the ECP. In the example in paragraph 5.2.4.2.1 above, the agency is the PBMA.

A - I&SA	G - GOCO AAP	P - ARDEC
C - COCO PLANT	H - USAEDH	Q - IBEA
D - HARRY DIAMOND LABORATORIES	M - PBMA	R - HQ AMCCOM
		S - GOGO ACTIVITY

Other agencies may be added as they are identified and a designator is assigned.

5.2.4.2.5 Sequence Number. The remaining four digits indicate the sequence number of the ECP. This allows a capability of 9999 ECPs to be assigned to any site by an agency. In the example above, the 0006 means that it is the sixth ECP originated by PBMA against the Radford site. It must be noted that these four digits are site, but not project, oriented.

5.2.4.3 Preparation of ECP Form. The following instructions are keyed to the block number on the ECP form.

5.2.4.3.1 Block 1a - ECP Number and Date. The originating agency assigns the ECP number. The same ECP number will be used on each partial baseline ECP until completion of the release. The ECP date is to be completed using the standard three-alpha abbreviation for the month. For example: 17 Jul 85 or 5 Sep 85. Once assigned, the date of the basic ECP remains constant and is not changed as the ECP is revised or corrected.

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5.2.4.3.2 Block 1b - Revision and Date. This block is left blank when the basic ECP is prepared. Enter R1 in this block the first time an ECP is revised after it has been distributed outside the originating agency. Enter R2 if the ECP is revised a second time and increase accordingly for subsequent revisions. An ECP may be revised only by the originating agency or the CCB. The date in the block is the date of the current revision. Revisions are used for changes that affect the technical content of an ECP.

5.2.4.3.3 Block 1c - Correction and Date. Leave blank when basic ECP is prepared. Enter C1 for first correction, and so forth. The date in the block is the date of the current correction. Corrections are used to correct errors and not to revise the technical content of an ECP. An ECP may be corrected only by the originating agency or the CCB. Corrections may be issued subsequent to revisions; thus an R1 becomes an R1C1. If necessary to revise the ECP again, it would become R2 with the correction block left blank as corrections are absorbed in subsequent revisions.

5.2.4.3.4 Block 1d - Type. Check the appropriate block to denote a change or baseline ECP. Baseline ECPs will be annotated under the word "baseline" (see below) to denote a complete or partial baselining action. Enter "C01" if the ECP represents a complete baseline release. If the ECP represents the first partial baseline release, it will be identified as "P01." The second partial baseline release will be "P02," etc. The final partial baseline release, which completes the action, will be identified by the prefix "C" and the appropriate sequence number, e.g., "C03." The two digits following the "P" or "C" prefix provide for up to 99 partial baseline releases.

d. TYPE	d. TYPE	d. TYPE
___ Change	___ Change	___ Change
___	___	___
___ Baseline	___ Baseline	___ Baseline
X C01	X P01	X C03

5.2.4.3.5 Block 1e - Priority. Check appropriate block.

5.2.4.3.6 Block 1f - Approval Need Date. Enter the date that CCB approval is required to minimize schedule impact. The date which allows less than normal processing time must be justified in block 4.

5.2.4.3.7 Block 1g - Category. Check the appropriate one single block that best identifies the category of the ECP. A cost reduction or design improvement must represent a significant savings/improvement to warrant approving a change to an active construction or equipment production contract.

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A - State-of-the-Art - A design change made as a result of an advance in the state-of-the-art, which will result in significant improvement in performance.

B - Baseline - Necesssary to establish a required baseline.

D - Design Improvement - Improvement in design which will result in material improvement in performance.

O - Operational Requirement - Necessary to correct a deficiency or resolve a conflict as required to achieve operational requirements.

P - Pollution Abatement - A change required as a result of changes in air or water pollution standards.

R - Cost Reduction - Any change, other than a VE change, submitted for the primary purpose of reducing costs.

S - Safety - Necessary to correct or avoid a safety hazard.

V - Value Engineering - A change resulting from a formal VE study, and submitted by either Government personnel or by a contractor under the VE provisions of the contract.

5.2.4.3.8 Block 2a - Sites. Enter name and location (state) of the plant(s) that will be affected.

5.2.4.3.9 Block 2b - Project No. Enter the seven-digit project control number as established by PBMA.

5.2.4.3.10 Block 2c - SP. Enter the two-digit number, when applicable, identifying major subproject work areas within a project as established by PBMA.

5.2.4.3.11 Block 2d - Commodity Division. Check appropriate single block.

5.2.4.3.12 Block 3 - Title. Give title of proposed change in 40 spaces or less. For a release ECP, the first word of the title must identify the baseline to be established, e.g., DESIGN B/L.

5.2.4.3.13 Block 4 - Description and Justification. For a change ECP, describe in detail the proposed engineering change; show new capability; provide defect, failure, or malfunction information. Answer questions: (1) what is accomplished by change? (2) what experience indicated need for change? (3) what is the impact if change is not approved or deferred? The justification for the ECP must be completed and provide all necessary information required for evaluation. If additional space is needed to describe the change, use the ECP continuation sheet. For a design baseline ECP, enter the contract number and date of award. List specifications and drawings by volumes and identify amendment revisions.

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5.2.4.3.14 Block 5a - Document Number and Title. Enter the number of each document affected by the proposed ECP. Facility drawings will be listed by file number. Enter NA for baseline ECP.

5.2.4.3.15 Block 5b - Current Revision. Insert the current revision symbol of the documentation to be changed.

5.2.4.3.16 Block 5c - Sheet/Page Numbers. Insert the sheet or page numbers primarily affected by the ECP.

5.2.4.3.17 Block 6a - Related ECP. Enter all ECPs (approved/pending) which relate to the proposed ECP.

5.2.4.3.18 Block 6b. Enter budget code.

5.2.4.3.19 Block 7a - Originator/Design Engineer. Enter the name, organization, and phone number of the individual who will be prepared to answer questions concerning the ECP.

5.2.4.3.20 Block 7b - Submitted By. The signature of an authorized representative of the agency initiating the ECP. All ECPs originated by the PBMA will be submitted by the Executive Secretary of the CCB.

5.2.5 ECP Processing Prior to CCB Action.

5.2.5.1 Distribution of ECPs. See Figure for flow diagram of a typical ECP. ECPs will be prepared on standard size formats utilizing black ink on white paper. They must be legible and capable of being satisfactorily microfilmed and reproduced. Attachments to ECPs must also be of the same quality, and sizes over 8½ x 14" may be prepared on translucent reproducible stock. To reduce processing time, the originating agency makes the initial direct distribution of the original and one copy to the Executive Secretary and two copies to each participating agency. Participating agencies will normally include all board members, the affected ammunition plant, and CE district as applicable. A simple transmittal form, no letter required, will be used to distribute ECPs. The transmittal must, however, indicate the entire distribution that has been made other than in-house copies in the originating agency. The distribution list for each CCB will be controlled by the Executive Secretary of that CCB.

5.2.5.2 ECP Review Period. The review period for each change ECP will depend upon the complexity of the proposed change and whether the ECP has been assigned a routine or an urgent priority. In general, a maximum of five working days review period after receipt of the ECP shall be allowed for urgent ECPs. Approximately 14 calender days shall be allowed for routine ECPs. This may be varied as directed by the CCB Chairman.

5.2.5.3 ECP Review and Coordination. See paragraph 5.2.1 for discussion of ECP review and coordination.

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5.2.5.4 Changes to ECPs. Any changes to an ECP after it has once been distributed outside the originating agency requires a revision or a correction to the ECP. Where the revision or correction is identified during the CCB meeting and noted in Block 4 of the concurrence sheet, it is not necessary to rewrite the ECP itself.

5.2.6 ECP Impact Sheet.

5.2.6.1 Impact Sheet Form. DA Form 5200-R (Production Base Engineering Change Proposal, Impact Sheet) will be prepared during the review period by each agency impacted by a proposed change ECP. DA Form 5200-R will be locally reproduced on 8½"x11" paper. Impact sheets will conform to the same quality requirements as ECPs. Completed impact sheets for routine ECPs shall be furnished to the Executive Secretary prior to the CCB meeting, the CCB member representing the impact agency shall be prepared to furnish a ROM estimate so that the members will know the general magnitude of the impact which they are considering. In this case, the impact sheet will be furnished at a subsequent meeting. If the impact is substantially above the ROM estimate, the Chairman may elect to reconsider the ECP if it has not already been implemented. Addition of the impact sheet does not in itself require a change in the revision or the correction block of the ECP form.

5.2.6.2 Preparation of ECP Impact Sheet. The following instructions are keyed to the block number on the form:

a. Block 1a - ECP Number

b. Block 1b - Revision

c. Block 1c - Correction

d. Block 1d - Title

e. Block 2a - Project Number and site affected

f. Block 2b - SP (Subproject)

The instructions for completion of these blocks are the same as those for the ECP Form.

5.2.6.2.2 Block 2c - Design Cost. Enter estimated cost of design effort required to incorporate change in design documentation.

5.2.6.2.3 Block 2d - Contract Cost. Enter estimated modification cost for projects under contract or which will be under contract when the ECP is implemented. Enter estimated change to the bid for projects where the ECP can be incorporated prior to contract award.

5.2.6.2.4 Block 2e - Other Cost. Enter estimated cost, excluding design, for reproduction, negotiation, inspection, and overhead associated with incorporating the approved ECP into the contract.

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FORM APPROVED
OMB NUMBER 0704–0188
NO EXPIRATION DATE

PRODUCTION BASE ENGINEERING CHANGE PROPOSAL (ECP)
IMPACT SHEET

1.	a. ECP NUMBER	b. REVISION	c. CORRECTION	d. ECP TITLE

			ESTIMATED COSTS				g. SCHEDULE EFFECT		
2.	a. PROJECT NO.	b. SP	c. DESIGN	d. CONTRACT	e. OTHER	f. TOTAL	NONE	MINOR	KEY
	TOTALS								

3. REMARKS

NAME

ORGANIZATION DATE

NOTE: THIS IMPACT ESTIMATE CONSIDERS ONLY THE INDIVIDUAL ECP, WITHOUT EVALUATION OF THE CUMULATIVE EFFECT OF SUCCESSIVE ECPs. WHERE MINOR SCHEDULE EFFECT IS CHECKED, NO SEPARATELY IDENTIFIABLE DELAY WILL RESULT FROM THE INDIVIDUAL ECP; HOWEVER, FOR SITES UNDER CONSTRUCTION OR PROCESS EQUIPMENT IN PRODUCTION, THE CUMULATIVE EFFECT OF A SERIES OF ECP MAY GREATLY EXCEED THE TOTAL OF INDIVIDUAL ESTIMATES OF TIME AND MONEY.

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5.2.6.2.5 Block 2f - Total Cost. Enter total of all columns. If savings, show in parenthesis to indicate decrease rather than increase in project cost.

5.2.6.2.6 Block 2g - Schedule Effect. Enter an "x" in the appropriate column:

a. None - Does not affect schedule.

b. Minor - Has no separately identifiable schedule impact, but taken together with other proj changes could result in setting back the completion date.

c. Key - Impacts a schedule milestone. A Key impact must be explained in the Remarks block and the magnitude identified. Should describe the add'l time estimated that the change will add to the construction time required.

5.2.6.2.7 Block 3 - Remarks. Enter any pertinent remarks. If additional authorization on funds are required, identify here.

5.2.7 CCB Procedures.

5.2.7.1 ECP Flow Diagram. See Figure for ECP flow diagram.

5.2.7.2 Scheduling CCBs. CCB meetings will be scheduled by the Executive Secretary and insofar as practical, will be held at regularly scheduled intervals. Urgent ECPs requiring action before a scheduled CCB meeting may be coordinated telephonically and then formalized at the next meeting. In case of extreme urgency, the PBMA Commander, or the Chairman of the CCB by direction of the Commander, may approve an ECP prior to review by the CCB members. In such instances, the ECP will be formalized at the next CCB meeting. The Executive Secretary will schedule incoming ECPs and distribute a tentative agenda approximately one week prior to each CCB meeting. This will be updated if necessary shortly before the meeting so that each member knows which ECPs are to be considered at the meeting.

5.2.7.3 ECP Review Prior to CCB Meeting. Prior to the CCB meeting, the Executive Secretary will review each ECP to insure that it is properly prepared. Corrections will be coordinated with the originating agency and noted in Block 4 of the concurrence sheet before it is presented to the CCB members for signature. If during the review period the ECP is determined to be incomplete or inaccurate, it will be removed from the agenda until it has been revised by the originator.

5.2.7.4 Split Boards. A split board decision at the PBMA CCB is referred by the Chairman to the PBMA Commander for resolution. After obtaining the Commander's decision, the Chairman then effects ECP disposition.

5.2.7.5 CCB Summary. Following the CCB meeting, the Executive Secretary prepares a summary list of the decision reached concerning each ECP on the agenda. A complete set of minutes is not required as significant remarks or instructions have been noted during the meeting in Block 7 of the concurrence sheet. The summary is distributed to the members together with copies of the concurrence sheets, revised ECP, and impact sheets not previously distributed. Enclosures larger than 8½ x 14" will not be redistributed with the ECP unless they have been revised other than as indicated in block 4 of the concurrence sheet.

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5.2.8 ECP Concurrence Sheet.

5.2.8.1 DA Form 5201-R. DA Form 5201-R will be used for concurrence.

5.2.8.2 Completion of Concurrence Sheet. The following instructions are keyed to the block numbers on the form.

5.2.8.2.1 1a - ECP Number, Date. Same as ECP.

5.2.8.2.2 1b - Project Number. Same as ECP with site affected.

5.2.8.2.3 1c - Revision, Date. Same as ECP.

5.2.8.2.4 1d - Correction, Date. Same as ECP.

5.2.8.2.5 1e - Revision, Date. Enter R1, R2..., when a change has been made to the ECP by the CCB. Enter date of revision. The CCB will not change revision block 1c.

5.2.8.2.6 1f - Correction, Date. Enter C1, C2..., when an editorial correction has been made to the ECP by the CCB. Enter date of correction. The CCB will not change correction block 1d.

5.2.8.2.7 2a - CCB. Enter PBMCCB for PBMA CCB or other designators as assigned by the PBMA.

5.2.8.2.8 2b - Meeting No. The sequential meeting number of the CCB: 5001 was the first meeting in CY85. The first digit indicates the last digit of the CY and the last three digits indicate the sequence number of the meeting during that CY.

5.2.8.2.9 2c - Date. Date of CCB meeting.

5.2.8.2.10 3a - Facility. Estimated increase or decrease in facility construction cost attributable to this ECP (obtain figures from ECP Impact Sheet(s)).

5.2.8.2.11 3b - Equipment. Estimated increase or decrease in equipment cost attributable to this ECP (obtain figures from ECP Impact Sheet(s)).

5.2.8.2.12 3c - Total. Estimated net increase or decrease in facility and equipment cost attributable to this ECP.

5.2.8.2.13 3d - Key Time. Is a key time impact estimated attributable to this ECP? Check appropriate block, "yes" or "no," (obtain from ECP Impact Sheet(s)).

5.2.8.2.14 4 - Modification(s) to Submitted ECP. If none, check "none." If modifications have been made by the CCB, check "as follows" and briefly describe modifications.

5.2.8.2.15 5 - Board Members' Position on ECP as Modified Above. Each CCB member including ad hoc members enter organization symbol, check "yes" or "no" and affix signature.

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5.2.8.2.16 6 - Action Taken on ECP. If the ECP is approved, with no further CCB action required, check "approved" and "completed action." If ECP is disapproved with no further CCB action required, check "disapproved" and "completed action." If ECP is approved as "URGENT," with rewrite and resubmission required, check "approved" and "cleanup ECP required." If ECP is deferred or withdrawn, check "other," and explain in block 7. If ECP is approved prior to receipt of cost impact form, check "approved" and "furnish impact."

5.2.8.2.17 7 - Remarks. Use for explanations of ECP/CCB action which does not constitute a modification to the submitted ECP.

5.2.9 Approval of Facility Changes.

5.2.9.1 Design Baseline Changes. It is not practical for the CCB to control all of the minor corrections and changes to the facility design in the construction contracts because of the sheer volume of such actions, which may run into the hundreds per week during the peak period of a multimillion dollar contract. The delay to the construction contract would also be prohibitive. The design baseline is therefore limited in scope to the items covered in the criteria baseline together with that portion of the design that has a critical relationship to the operation of or interface with the equipment. For example, process piping to certain items of equipment is critical and will not be rerouted or otherwise redesigned without an approved ECP. All interface changes, either physical, functional, or environmental, will be by ECP. The typical minor noninterface design change will be reviewed and processed through normal CE management channels rather than by formal CCB procedures. Informal coordination with the Plant Commander will be accomplished to minimize the possibility of issuing a non-ECP facility change that inadvertently might impact the plant operation. Copies of incoming contractor correspondence that may generate a minor non-ECP facility change will be furnished by the CE resident engineer to the plant commander for his review. If his review indicates that the operation of the plant might be affected by the CE action on the correspondence, then he will notify the CE resident engineer within 48 hours after receipt of the letter. Considerable judgment is required on the part of the construction agency to insure that ECPs are in fact processed for changes that affect the using agency. The approval for non-ECP changes remains with the contracting officer. Such changes are beyond the scope of this plan and will not be "tracked" in the formal configuration accounting system. Each implementing agency will continue to use its own system to record such changes. In case of doubt as to whether an ECP is required, proposed facility changes will be referred to the configuration management staff of USAEDH for determination.

5.2.9.2 Requirement for ECP. Any change which is of sufficient magnitude that it appears likely to create an identifiable slippage in an established milestone and any change which requires action to reprogram funds will be processed by ECP regardless of whether or not an interface is affected. A logic diagram for determination of the ECP requirement is shown on Figure 4 of this standard.

5.2.10 Approval of Equipment Changes.

5.2.10.1 Equipment Changes. An ECP will be required for equipment changes if any of the following conditions are met:

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a. The change affects previously baselined documentation.

b. The change affects an interface with the facility.

c. The change delays the contract schedule by 30 days or more.

d. The change affects the cost by $10,000 or more (or as established by the Commander PBMA) or requires action to reprogram funds.

e. The change affects the safety of the equipment or an operator.

5.2.10.2 Editorial Changes. Editorial changes which do not affect cost, schedule, interface, or safety are exempt from ECP procedures.

5.2.10.3 Determination of ECP Requirement. A logic diagram for determination of the ECP requirements is shown on Figure 4 and 5 in this standard. In cases of doubt as to whether an ECP is required, proposed equipment changes will be referred to the Executive Secretary of the PBMA CCB.

5.2.11 ECP Implementation. To assist in tracking the implementation and cost factors of approved facility ECPs into ongoing construction contracts, each change order package shall separately identify ECPs which are incorporated in the package. As a standard procedure, the contract modification shall list each ECP as a separate line item with its actual negotiated cost shown.

5.2.12 ECP Feedback. Procuring Contracting Officer shall provide feedback data as expeditiously as possible to the Executive Secretary of the PBMA CCB. The required feedback data consists of:

a. ECP Number.

b. Contract Number.

c. Change order number and date if applicable. (Note: Change order data submission must be followed-up by contract modification data submission).

d. Contract modification number and date.

e. Negotiated/actual cost.

Feedback data should be transmitted on a locally developed transmittal-type form letter with authorization for lower level management approval. The feedback process should be kept simple and economical, yet responsive.

5.3 Configuration Accounting.

5.3.1 General. Each design agency participating in the program presently has an adequate procedure in effect to record the revision status of the design documentation controlled by that agency. Existing procedures will continue unchanged except as they may require modification to comply with the remainder of this section.

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5.3.2 Drawing Revisions. Where baselined drawings are revised to incorporate an approved ECP, the ECP number shall be shown in the revision block of each drawing revised to implement the ECP. No special procedure is required for ECPs incorporated prior to baseline of the drawing.

5.3.3 Specification Changes. Changes and revisions to baselined specifications shall be in accordance with paragraph 3.3 of MIL-STD-490. A specification change ECP, unless preliminary or broad in scope and not suitable for definitization, shall reference each change and include the exact word change in paragraphs, drawings, or other content. After receipt of the approved ECP, the originator of the specification shall effect formal recording and publication of the change. A Specification Change Notice (SCN) with the changed specification pages shall be distributed to all holders of the specification.

5.3.4 ECP Status Reporting. USAEDH will, in support of PBMA, maintain a computerized ECP data base capable of providing various status reports for ECPs that have been submitted to the CCB. To support this reporting requirement, construction and procurement offices that implement approved ECPs shall furnish USAEDH with implementation information for each ECP.

5.3.5 Repository. A central documentation repository will not be established. Record copies of each ECP will be available from the Executive Secretary of the CCB approving the ECP. Each design agency will maintain the record copy of design documentation controlled by that agency and will distribute copies to participating agencies for review or information in compliance with established distribution lists.

6: NOTES

6.1 Intended Use. This standard contains requirements for the application of Configuration Management to the modernization of the Army Armaments, Munitions, and Chemical Production Base Modernization.

6.2 Data Requirements List and Cross Reference. When this standard is used in an acquisition which incorporates a DD Form 1423, (Contract Data Requirements List) (CDRL), the data requirements identified below shall be developed as specified by an approved DD Form 1664 (Data Item Description) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, sub part 27.475.1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this standard is cited in the following paragraphs.

	TASK PARAGRAPH	DATA REQUIREMENTS	APPLICABLE DID	OPTIONS
a.	5.1.2.1	Project Development Brochure	DI-CMAN-80396	-----
b.	5.1.3.2	Equipment Functional Criteria Specification	DI-CMAN-80397	-----

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c.	5.1.3.3	Equipment Technical Data Package Specification	DI-CMAN-80398	---
d.	5.1.4.1	List of Documentation	DI-CMAN-80399	---
e.	5.2.1.3 & 5.2.4	Engineering Change Proposal	DI-CMAN-80400	---

(Copies of DIDs required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer).

6.3 Subject Term (Key Word) Listing.

Army Production Base
Baseline
Concept Baseline
Configuration
Configuration Accounting
Configuration Control
Configuration Control Board
Configuration Identification
Configuration Item
Configuration Management
Criteria Baseline
Design Baseline
Engineering Change
Engineering Change Proposal
Equipment Functional Criteria Specification
Executive Secretary
Field Configuration Control Board
Functional Baseline
Modernization and Expansion
Operational Baseline
Process Baseline
Routing ECP
Specification
Technical Baseline
Technical Data Package
Value Engineering

6.4 Changes from previous issue. Vertical lines or asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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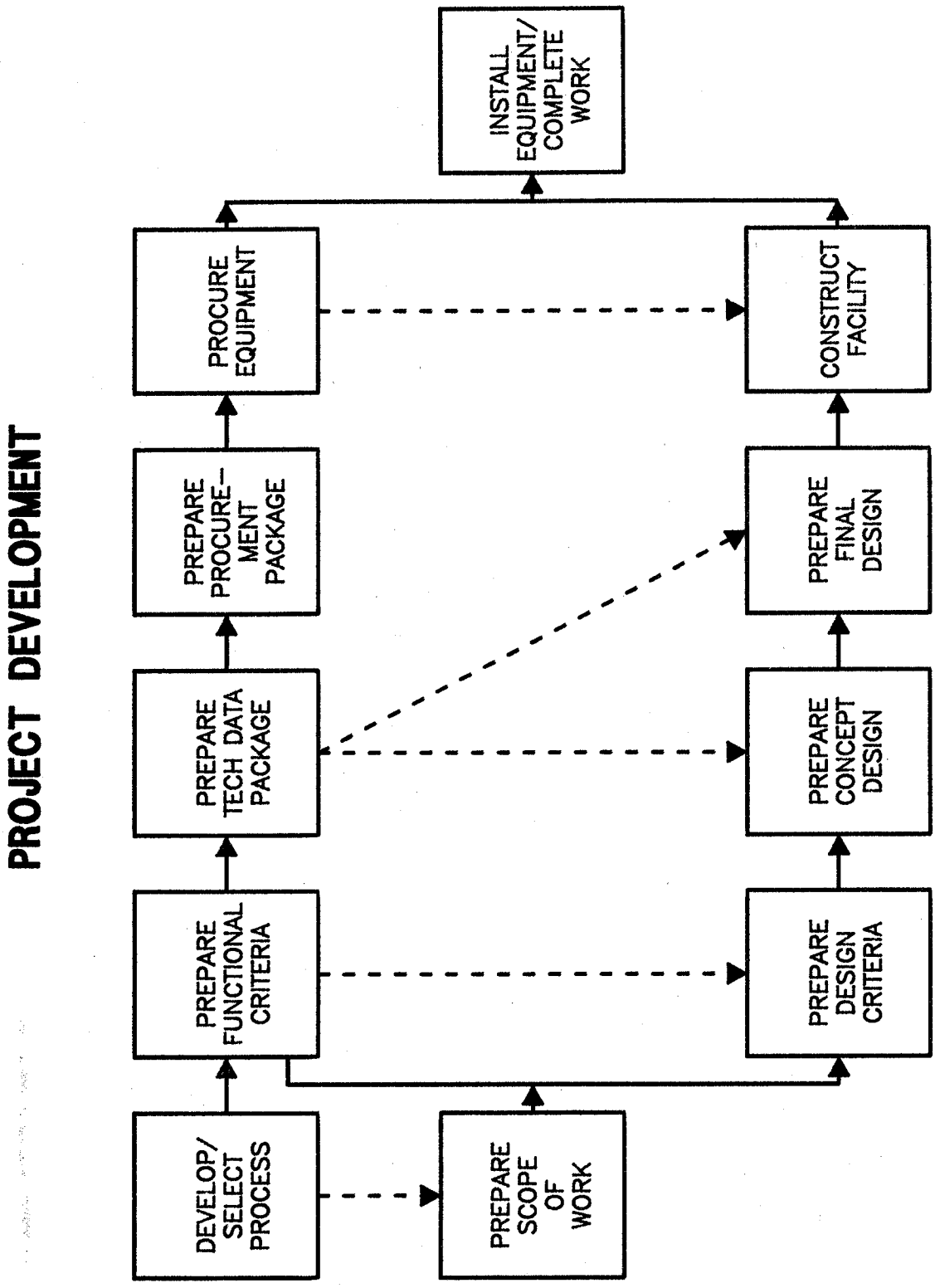


FIGURE 1. PROJECT DEVELOPMENT

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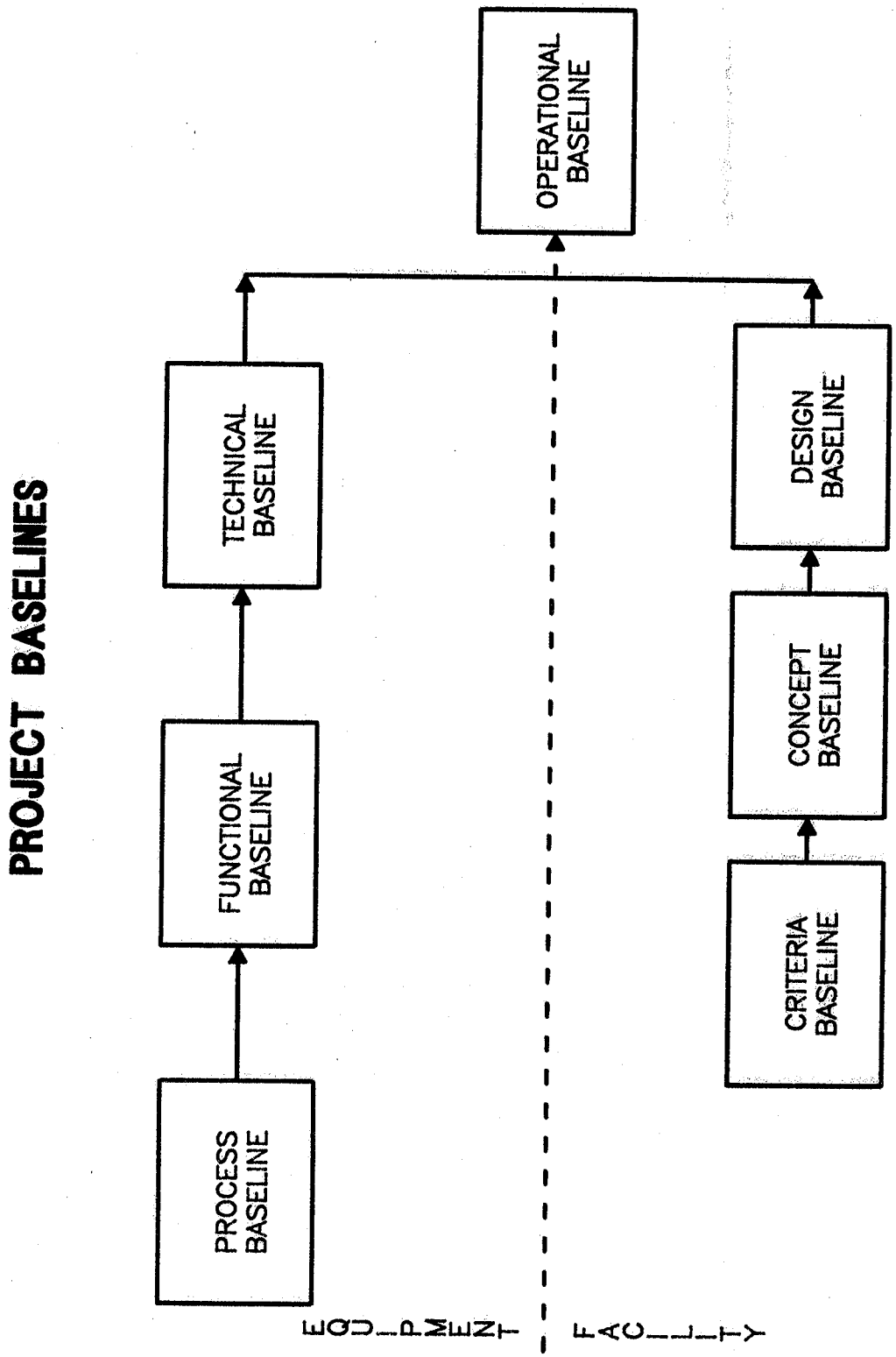


FIGURE 2. PROJECT BASELINES

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ECP FLOW DIAGRAM

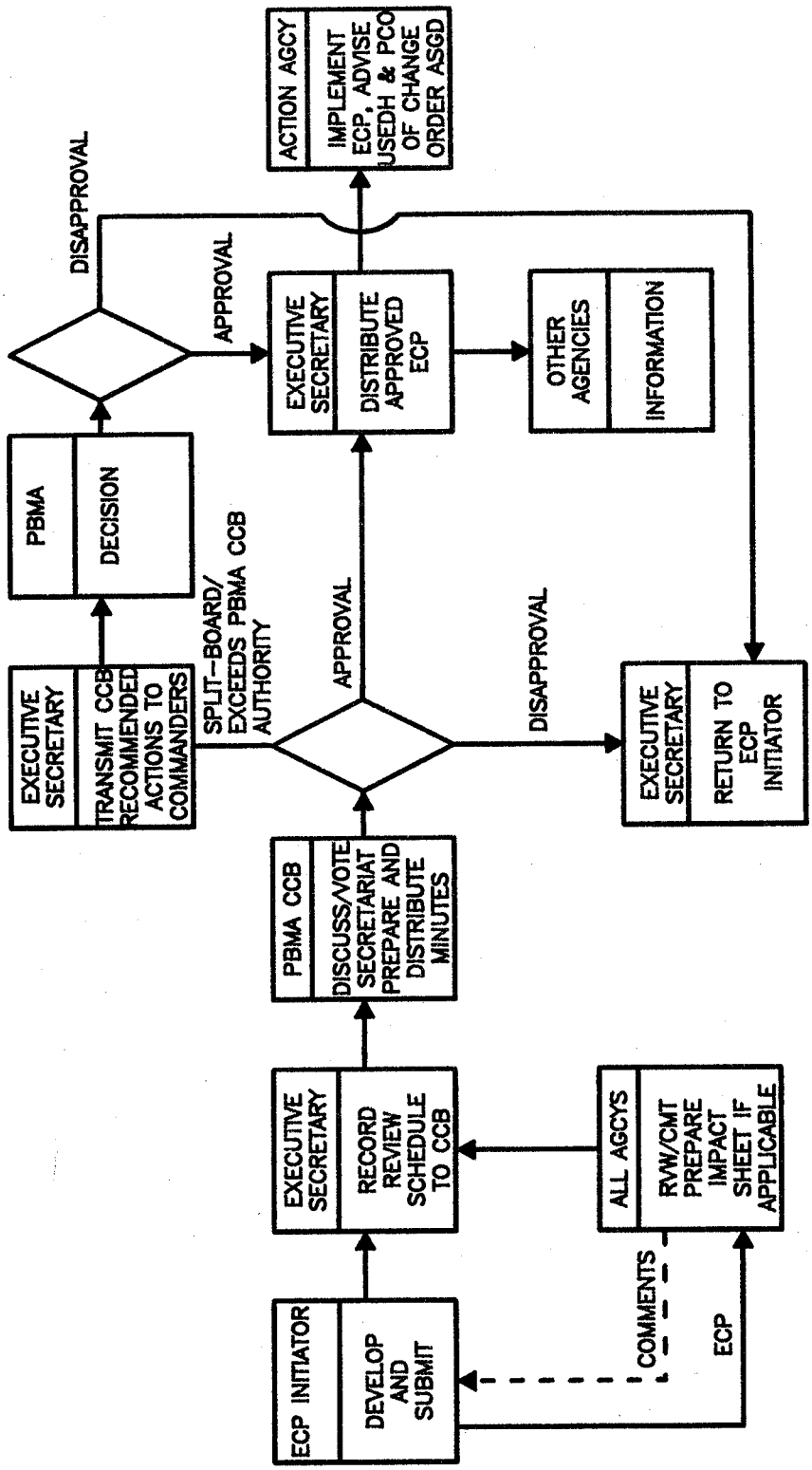


FIGURE 3. ECP FLOW DIAGRAM

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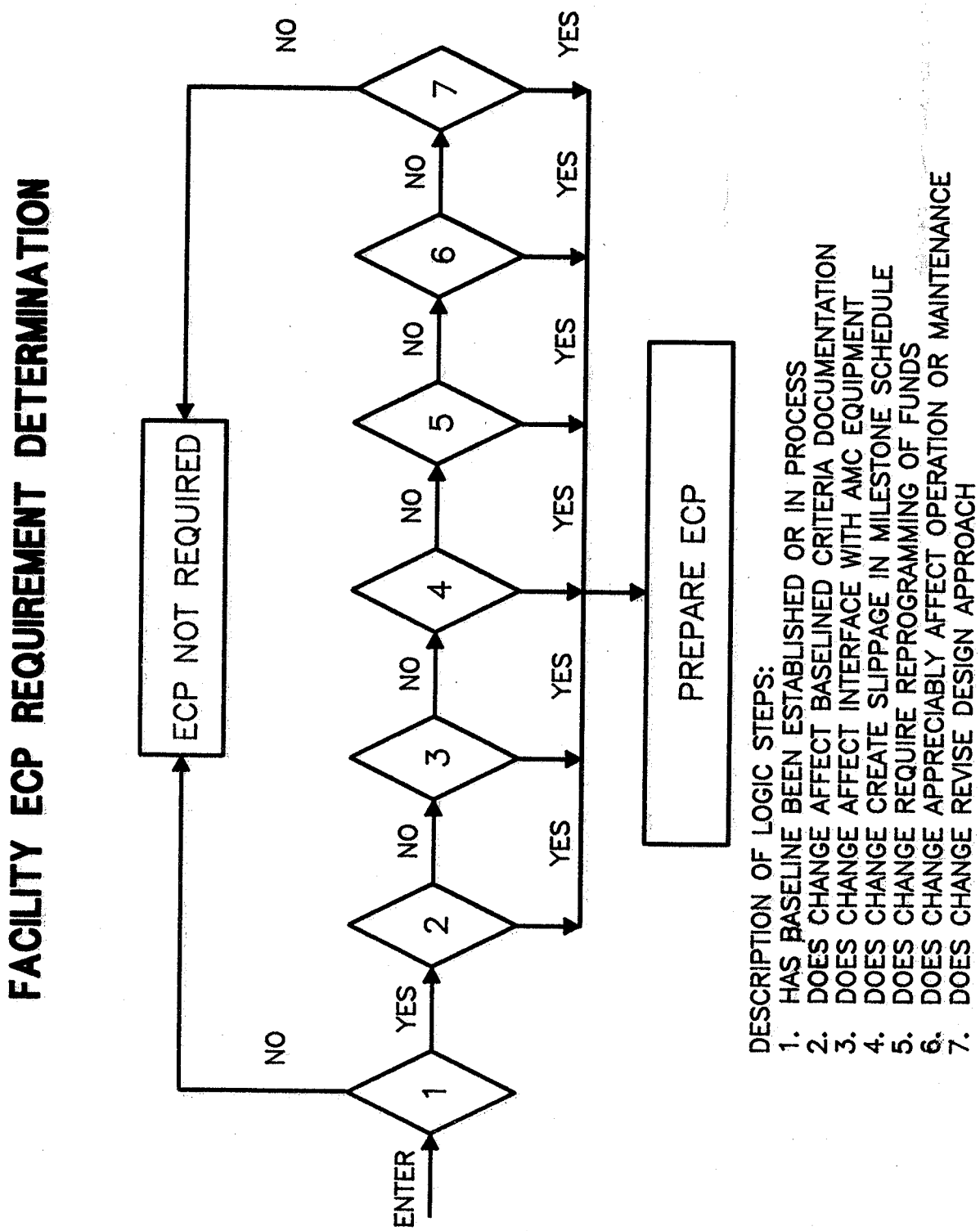


FIGURE 4. FACILITY ECP REQUIREMENT DETERMINATION

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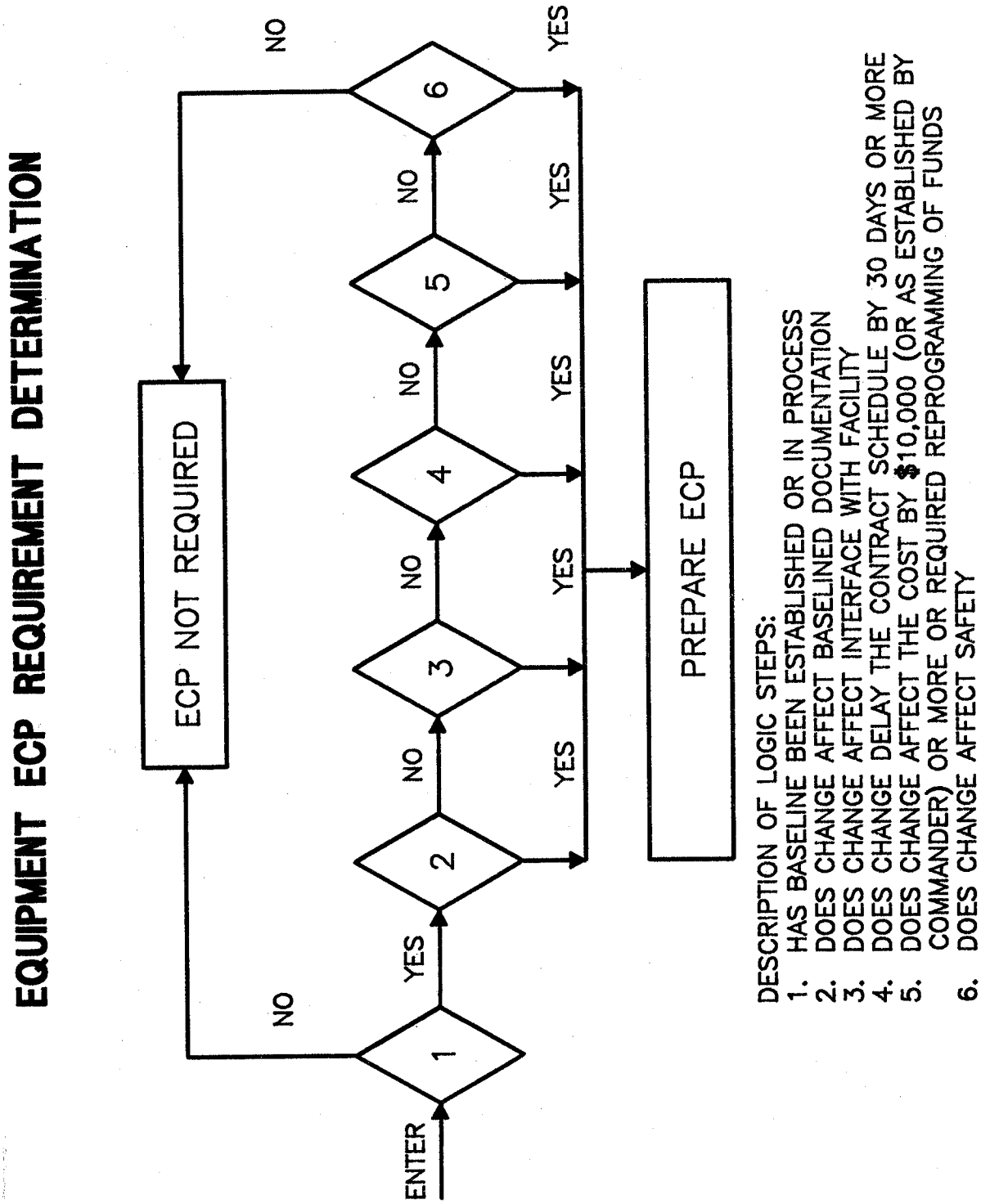


FIGURE 5. EQUIPMENT ECP REQUIREMENT DETERMINATION

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EQUIPMENT FUNCTIONAL CRITERIA SPECIFICATION

FOR

(PROJECT TITLE)

SPECIFICATION FS (Project No.)

REV

AMC Project XXXXXXX

(Date)

Preparing Organization

(Name)
(Code Identification)

FIGURE 6. Equipment Functional Criteria Specification

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TABLE OF CONTENTS

Section	
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3.1.1	General Description
3.1.2	Capability
3.1.3	Manufacturing Diagrams
3.1.4	Interface Definition
3.1.5	Government Furnished Property
3.1.6	Organizational Concepts
3.2	Characteristics
3.2.1	Performance Characteristics
3.2.2	Physical Characteristics
3.2.3	Reliability
3.2.4	Maintainability
3.2.5	Availability
3.2.6	Environmental Conditions
3.3	Design and Construction
3.3.1	Nameplates and Marking

FIGURE 7. Table of Contents

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3.3.2	Interchangeability
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3.4	Process Characteristics/Configuration Items
3.5	Logistics
3.5.1	Support
3.5.2	Facilities
3.5.3	Utility Consumption
3.5.4	Personnel
4.0	Test/Verification

FIGURE 7. Table of Contents - Continued

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EQUIPMENT TECHNICAL DATA PACKAGE SPECIFICATION

FOR

(EQUIPMENT NOMENCLATURE)

SPECIFICATION TS (PROJECT NO) - (SEQ NO)

AMC PROJECT XXXXXXX

(DATE)

Preparing Organization

(Name

Code Identification)

FIGURE 8. Equipment Technical Data Package Specification

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TABLE OF CONTENTS

Section	
1.	Scope
2.	Applicable Documents
2.1	Government Documents
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3.1.4	Government Furnished Property List
3.2	Characteristics
3.2.1	Performance Characteristics
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3.2.3	Reliability
3.2.4	Maintainability
3.2.5	Availability
3.2.6	Environmental Conditions
3.2.7	Transportability
3.3	Design and Construction
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3.3.2	Tooling
3.3.3	Identification and Marking

FIGURE 9. Table of Contents

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3.3.4	Interchangeability
3.3.5	Safety
3.3.6	Human Performance/Human Engineering
3.3.7	Standards of Manufacture
3.3.8	Facilities
3.3.9	Utility Consumption
3.4	Major Component Characteristics
4.	Quality Assurance Provisions
4.1	General
4.2	Quality Conformance
5.	Preparation for Delivery

FIGURE 9. Table of Contents - Continued

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APPENDIX A

EQUIPMENT FUNCTIONAL CRITERIA SPECIFICATION

10. GENERAL

10.1 SCOPE

10.1.1 Format. This appendix establishes the format and contents for the equipment functional criteria specification, hereafter referred to as the functional specification.

10.1.2 Purpose. The purpose of this appendix is to establish uniform practices for specification preparation, to ensure the inclusion of essential requirements, and to aid in the use and analysis of specification content.

10.1.3 Scope of Functional Specifications. The functional specification states the technical and mission requirements of a modernization or expansion project as an entity, allocates requirements to functional areas (processes), identifies interfaces, and specifies individual pieces of equipment which are to be configuration items (i.e., critical to the process, having a high cost, separately procurable, etc.).

10.1.4 Basis of Functional Specification. The functional specification normally is based on the results of an MMTE effort, or an exploratory preliminary design period (omnibus) of feasibility studies and analyses.

10.1.5 Updating Functional Specification. The functional specification is maintained current during the entire modernization/expansion effort.

20. REFERENCED DOCUMENTS

MIL-STD-490
Cataloging HDBK H4

30. CHANGES.

30.1. Changes to the Equipment Functional Criteria Specification. Shall be in accordance with MIL-STD-490 paragraph 3.3 with the following exceptions:

a. All SCNs shall be related to an ECP.

b. Paragraph 3.3.2.1.1 (Proposed SCN) is not applicable. The SCN shall be an enclosure to the related ECP.

40. GENERAL REQUIREMENTS..

40.1 Identification of Functional Specification. The functional specification shall be identified with the letters FS followed by the seven-digit project number. The month and year of preparation shall appear on the cover sheet.

40.2 Identification Number and Location. The identification number, including prefix letters, shall appear on each page; it shall be located at the top of the page opposite the binding edge. Those pages changed by SCN shall have the date of the SCN below the identification number - see paragraph 3.3.2.4.1 of MIL-STD-490.

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APPENDIX A

40.3 Revision Letters. Revision letters, starting with "A" for the first revision and assigned alphabetically for each succeeding revision, shall follow the specification number. Letters such as I, O, and Z, which can be confused with numerals shall not be used.

40.4 Cover Sheet and Contents. Each functional specification shall have a cover sheet and Table of Contents prepared as shown in Figures 6 and 7, respectively. The cover sheet shall contain the following information:

a. Title - This is the title of the AMC project such as, 105mm Projectile Manufacturing Facility Modernization Project.

b. Specification Number - See paragraph 20.3.1.

c. Project Number - AMC Project Number (7 digits).

d. Date - Month and year the specification was prepared.

e. Revision Level - This information shall not be shown for the original issue of the specification.

f. Preparing Organization - The AAP or Commodity Center which prepared the functional specification and its code identification as listed in Cataloging Handbook H-4.

40.5 Detail Requirements. Requirements for sections and paragraphs are listed below. Requirements that do not properly fall under a paragraph number or title may be added as additional paragraphs in the appropriate section. The notation "Not Applicable" shall be entered after each paragraph number and title that is not applicable. Subordinate paragraph headings may be added under the most suitable major paragraph heading in the outline prescribed.

Section 1. Scope. The content of Section 1 shall be as defined in the following example:

Example:

1. Scope

1.1 This equipment functional criteria specification establishes the necessary design and performance characteristics of the manufacturing line for the (insert project title) effort.

Section 2. Applicable Documents. All and only those documents referenced in Section 3, 4, and appendices of the specification shall be listed in Section 2. If numerous, Section 2 may reference an appendix. References shall be confined to documents currently available at the time of issuance or revision of the specification. Figures bound integrally with the specification shall not be listed in Section 2.

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APPENDIX A

Paragraph 2.1 - Government Documents. Government SPECIFICATIONS, STANDARDS, DRAWINGS, and other PUBLICATIONS intended to be made available shall be listed under the appropriate preceding heading and in alphabetical numerical order in individual groups. These listings shall be included under a paragraph similar to the following:

Example:

2.1. Government Documents. The following documents of the issue in effect on the date or revision of this specification form a part of this specification to the extent specified herein.

Government documents shall be listed in the following order:

SPECIFICATIONS

Federal

Military

Other Government Activity

STANDARDS

Federal

Military

Other Government Activity

DRAWINGS: (where detailed drawings referred to in the specification are listed on an assembly drawing, it is only necessary to list the assembly drawing).

OTHER PUBLICATIONS

Manuals

Regulations

Handbook

Bulletins, etc.

Paragraph 2.2 - Nongovernment documents. Nongovernment documents shall be listed in appropriate order under a paragraph similar to the following:

Example:

2.2 Nongovernment Documents: The following documents form a part of this specification herein. Nongovernment documents shall be listed in the following order:

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SPECIFICATION

STANDARDS

DRAWINGS

OTHER PUBLICATIONS

Section 3. Requirements. This section shall contain the following:

a. The performance and design requirements for the manufacturing line as a result of the modernization/expansion effort.

b. The performance requirements related to manning, operating, maintaining, and logistically supporting the manufacturing line to the extent these requirements define or constrain design of process equipment.

c. The definition of the principal interfaces between the manufacturing line being specified, the facility, and the item produced.

d. The individual processes of the manufacturing and the principal interfaces between each process.

e. The allocation of performance to and the specific design constraints peculiar to each process.

f. The identification and use of major Government-furnished property to be designed into and delivered with other equipment as an entity and an integral part of system capability.

Paragraph 3.1 - Manufacturing Line Definition.

Paragraph 3.1.1 - General Description. This paragraph shall contain a brief description of the manufacturing line.

Paragraph 3.1.2 - Capability. This paragraph shall delineate the overall production capability of the manufacturing line and shall show (directly or by reference) the mobility requirement(s) that can be supported.

Paragraph 3.1.3 - Manufacturing Line Diagrams. This paragraph shall incorporate (directly or by reference) the manufacturing line functional diagrams. This paragraph shall include the top-level functional flow diagram(s) of the manufacturing line, to the level of detail required to identify all processes or operations of the line. System-level layout drawings or other graphic portrayal which establish the general relationship of processes or operations shall be included herein. Included shall be a list of the processes or operations keyed to the top-level functional diagram.

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APPENDIX A

Paragraph 3.1.4 - Interface definition. This paragraph shall contain subparagraphs which cover the functional and physical interfaces.

Paragraph 3.1.4.1 shall discuss the interfaces between the manufacturing line and other lines with which it must be compatible.

Paragraph 3.1.4.2 shall discuss the interface between the manufacturing line and the facilities. This paragraph shall address the utilities required. (Utility consumption shall be addressed in paragraph 3.6.3). Reference may be made to the Project Development Brochure if it has been prepared.

Paragraph 3.1.4.3 shall delineate the interfaces between the processes or operations within the manufacturing line in general terms. These interfaces shall be addressed in detail in paragraph 3.5.

Paragraph 3.1.5 - Government furnished property list. This paragraph shall list the Government furnished equipment which the manufacturing line shall be designed to incorporate. This list shall identify the equipment by reference to its nomenclature, specification number, and part number. If the list is extensive, it may be included in an appendix or a separate document which shall be referenced in this paragraph.

Paragraph 3.1.6 - Organizational Concepts. This paragraph shall address the relationship of this manufacturing line in relationship to related manufacturing lines and the anticipated replication of the line such as the number of installations and their operating locations.

Paragraph 3.2 - Characteristics.

Paragraph 3.2.1 - Performance Characteristics. This paragraph shall contain performance characteristics including such considerations as quantitative criteria covering endurance capabilities of the manufacturing line while supporting the required mobilization rate, including minimum total life expectancy. Other essential aspects of the manufacturing line which cannot be more appropriately located under another heading in this outline, shall be stated.

Paragraph 3.2.2 - Physical Characteristics. This paragraph shall cover the following as applicable:

a. Dimensional and cube limitations for the manufacturing line, operator station layout, ingress, egress, and access for maintenance.

b. Security criteria.

c. Vulnerability factors.

d. Health criteria, including consideration of adverse explosive, mechanical, and biological effects. Included in these criteria are the toxological effects of the item being produced. OSHA requirements may be referenced in this paragraph.

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APPENDIX A

Paragraph 3.2.3 - Reliability. Reliability shall be stated in quantitative terms, defining the conditions under which the reliability requirements are to be met. This paragraph may include a reliability apportionment of reliability values assigned to processes for their share in achieving desired manufacturing line reliability.

Paragraph 3.2.4 - Maintainability. This paragraph shall specify the quantitative maintainability requirements. The requirements shall apply to maintenance in the planned maintenance and support environment and shall be stated in quantitative terms. Examples are:

a. Time (e.g., mean and maximum downtime, mean and maximum times to repair, mean time between maintenance actions).

b. Rate (e.g., maintenance man-hours per specific maintenance action).

c. Maintenance complexity (e.g., number of people and skill levels, variety of support equipment).

d. Maintenance action indices (e.g., maintenance costs per operating hour).

Paragraph 3.2.5 - Availability. The degree to which the manufacturing line shall be in an operable state at the start of production where the production is called for at an unknown point in time. This requirement shall not address time required to commence production after a line has been "laid away."

Paragraph 3.2.6 - Environmental conditions. This paragraph shall include the impact of the manufacturing line on the natural environment (including the generation of electromagnetic radiation).

Paragraph 3.3 - Design and Construction.

Paragraph 3.3.1 - Nameplates and Marking. This paragraph shall contain requirements for nameplates, part marking serial and lot nubmer markings, and all other identifying markings required for the equipment and its parts. Requirements shall be stated in general terms and reference made to existing standards on the content and application of such markings.

Paragraph 3.3.2 - Interchangeability. This paragraph shall specify the requirements for the equipment to be interchangeable and replaceable. Entries in this paragraph are for the purpose of establishing a condition of design and are not to define the conditions of interchangeability that are required by the assignment of a part number.

Paragraph 3.3.3 - Safety. This paragraph shall specify those safety requirements which are basic to the design of the manufacturing line, with respect to equipment characteristics, methods of operation, and environmental influences, and which prevent

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personnel injury and equipment degradation, without degrading operational capability. These requirements include such things as: restricting the use of dangerous materials when possible; classifying explosives for purposes of shipping, handling, and storing; abort/escape provisions; gas detection and warning devices; grounding of electrical systems; cleanliness and decontamination; explosive proofing, etc.

Paragraph 3.3.4 - Human Performance/Human Engineering. Human engineering requirements for the manufacturing line should be specified herein and applicable documents included by reference. This paragraph should also specify any special or unique requirements. Included should be those specific areas, stations, or equipment which would require concentrated human engineering attention due to the sensitivity or criticality of the task, i.e., those areas where the effects of human error would be particularly serious.

Paragraph 3.4 - Process or Operation Characteristics/Configuration Items. This paragraph shall include a subparagraph for each process listed in paragraph 3.1.3 of the functional specification. If this list is extensive, it shall be included in an appendix which shall be referenced here. The following information shall be in each subparagraph for process or operation:

a. Identification of the process or operation to the top-level flow diagram.

b. Functional and physical interfaces.

c. Functional and flow diagrams, as needed to identify, define the parameters of and within the process.

d. Performance and physical characteristics of the process.

e. Identification of the equipment used in the process and identification of the agency who will procure and install the equipment (DARCOM or CE). The equipment listed here is that which is defined as a configuration item because of its criticality to the process, high cost, procurement pattern/lead time, etc.

Paragraph 3.5 - Logistics.

Paragraph 3.5.1 - Support. This paragraph shall specify any unique impacts on the existing supply system for the procurement of spare/repair parts.

Paragraph 3.5.2 - Facilities. This paragraph shall incorporate (directly or by reference) all related facility documentation such as the Project Development Brochure, drawings, etc. This data may be included in an appendix and referenced here.

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APPENDIX A

Paragraph 3.5.3 - Utility Consumption. This paragraph will list the utility requirements for the manufacturing line as an entity. (If this information is contained in other documentation it may be referenced here). Utilities for heating, lighting, and sanitation within the facility shall not be listed here. A table should be provided (in an appendix if necessary) to summarize utility requirements by process. The following should be covered as a minimum:

a. Electrical Power

b. Fuel Oil

c. Gas

d. Water

e. Steam

The total requirement shall be listed as well as data such as pressure, flow rate recirculation, etc.

Paragraph 3.5.4 - Personnel. Paragraph shall specify personnel requirements which must be integrated into the manufacturing line requirements. It shall include number and type of personnel required for the designed mobilization rate(s).

Section 4. Test/Verification. Requirements for formal test/verification of the manufacturing line performance and design characteristics and operability shall be specified in this paragraph. Tests/verification specified herein shall include configuration item design evaluation, integrated manufacturing line evaluation, and operability capability verification. Subparagraphs under this section shall include:

a. Reliability testing with respect to manufacturing line reliability. Requirements shall be specified for collection and recording of data during all testing which is part of the reliability analysis.

b. Engineering evaluation and test requirements in direct support of design and development activity.

c. Qualification testing of configuration items and critical items (must include all items defined in paragraph 3.4).

d. Installation testing and checkout.

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APPENDIX B

EQUIPMENT TECHNICAL DATA PACKAGE (TDP) SPECIFICATION

10. GENERAL.

10.1 Scope. This appendix establishes the format and contents for the equipment TDP specification hereafter referred to as the technical specification.

10.2 Purpose. The purpose of this appendix is to establish uniform practices for specification preparation, to ensure the inclusion of essential requirements, and to aid in the use and analysis of specification content.

10.3 Content. The technical specification states (1) the complete performance requirements of the equipment for the intended use and (2) necessary inerface and interchangeability characteristics. It covers form, fit, and function. Complete performance requirements include all essential functional requirements under operating conditions necessary to support the specified mobilization requirements.

10.4 Application. The technical specification is applicable to the items below the total manufacturing line level and is prepared for procurement of equipment through specification of primarily function (performance) requirements.

10.5 Basis of Need. The technical specification is normally based on the refinement of requirements stated in the equipment functional criteria specification. A technical specification shall be written for each configuration item identified in paragraph 3.4 of the equipment functional criteria specification.

10.6 Updating. The technical specification is maintained current during the entire modernization/expansion effort.

20. REFERENCED DOCUMENTS.

MIL-STD-490
Cataloging HNDBK H4

30. CHANGES.

30.1 Basis of Changes. Changes to the technical specification shall be in accordance with MIL-STD-490 paragraph 3.3 with the following exceptions:

a. All SCNs shall be related to an ECP.

b. Paragraph 3.3.2.1.1 (Proposed SCN) is not applicable. The SCN shall be an inclosure to the related ECP.

40. GENERAL REQUIREMENTS.

40.1 Identification of Technical Specification. The technical specification shall be identified with the letters TS followed by the seven-digit project number followed by a three-digit sequence number. The three-digit sequence number shall start with 001 for the first specification and continue in numerical sequence for each additional specification.

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20.3.2 Identification Number. The identification number (including letter prefix) shall appear on each page; it shall be located at the top of the page opposite the binding edge. Those pages changed by SCN shall have the date of the SCN below the identification number - see paragraph 3.3.2.4.1 of MIL-STD-490.

20.3.3 Revisions. Revision letters, starting with "A" for the first revision and assigned alphabetically for each succeeding revision, shall follow the specification number. Letters such as I, O, Z which can be confused with numerals shall not be used.

20.3.4 Cover Sheet and Contents. Each technical specification shall have a cover sheet and Table of Contents prepared as shown in Figures 8 and 9, respectively. The cover sheet shall contain the following information:

a. Nomenclature of the equipment for which the specification was prepared such as - Nosing Press for 105mm Projectile Manufacturing facility.

b. Specification Number - See paragraph C-3-1.

c. Revision Level - This information shall not be shown for the original issue of the specification.

d. Project Number - AMC seven-digit project number.

e. Date - Month and year the specification was prepared.

f. Preparing Organization - The AAP or Commodity Center which prepared the technical specification and its code identification as listed in Cataloging Handbook H-4.

20.4 Detail Requirements.

20.4.1 Sections and Paragraphs. Requirements for sections and paragraphs are listed below. Requirements that do not properly fall under a paragraph number or title may be added as additional paragraphs in the appropriate section. The notation "Not Applicable" shall be entered after each paragraph number and title that is not applicable. Subordinate paragraph headings may be added under the most suitable major paragraph heading in the outline prescribed.

Section 1. Scope. The content of Section 1 shall be as defined in the following example:

Example:

1. Scope

1.1 This equipment technical data package specification establishes the performance, design, test, manufacture, and acceptance for the (insert equipment title) for the (insert project title).

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APPENDIX B

Section 2. Applicable Documents. All and only those documents referenced in Sections 3, 4, and 5 and appendixes of the specification shall be listed in Section 2. If numerous, Section 2 may reference an appendix. References shall be confined to documents currently available at the time of issuance or revision of the specification. Figures bound integrally with the specification shall not be listed in Section 2.

Paragraph 2.1 - Government Documents. Government SPECIFICATION, STANDARDS, DRAWINGS, and other PUBLICATIONS intended to be made available shall be listed under the appropriate preceding heading and in alphabetical-numerical order in individual groups. These listings shall be included under paragraph similar to the following:

Example:

2.1 Government Documents. The following documents of the issue in effect on the date of invitation for bids or request for proposal shall form a part of this specification to the extent specified herein:

Government documents shall be listed in the following order:

SPECIFICATIONS

Federal
Military
Other Government Activity

STANDARDS

Federal
Military
Other Government Activity

DRAWINGS (Where detailed drawings referred to in the specification are listed on an assembly drawing, it is only necessary to list the assembly drawing).

OTHER PUBLICATIONS

Manuals
Regulations
Handbooks
Bulletins, etc.

(List source for all documents not available through normal Government stocking activities).

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APPENDIX B

Section 3. Requirements.

Paragraph 3.1 - Item Definition. This paragraph shall contain a comprehensive definition of the functional characteristics of the item which is covered by this specification. The relationship of the equipment to the manufacturing line shall be defined, (i.e., define the process which the item supports; if there is more than one configuration item supporting a process, then identification of the other equipment shall be referenced here).

Paragraph 3.1.1. Item Diagrams. This paragraph shall incorporate functional schematic and flow diagrams. Applicable layout drawing or other graphic portrayals which establish the general relationship of major components of the item shall be included.

Paragraph 3.1.2. Interface Definition. This paragraph shall cover the functional and physical interfaces between (a) the item and other items; (b) the major components within the item, and (c) the item and the facility. The functional interfaces shall be specified in quantitative terms of input voltages, water pressure and flow, steam pressure, flow, conveyor speed, input/output flow rates, etc. Where interfaces differ due to a change in operational mode (rate), the requirements shall be specified in a manner which identifies specific functional interface requirements with each different mode (rate). Physical interface relationships shall be expressed in terms of dimensions with tolerances. This paragraph shall incorporate, either directly or by reference, interface control drawings and other engineering data as necessary to define all functinal and physical interfaces required to make the item compatible with other items and to make its major components compatible within the item.

Paragraph 3.1.3. Major Component List. This paragraph shall include a complete list of all major components which comprise the item.

Paragraph 3.1.4. Government Furnished Property List. This paragraph shall list the Government-furnished property which is required for item fabrication. This list shall identify the property by reference to its nomenclature, specification number, part number, and other pertinent identifiers.

Paragraph 3.2. Characteristics.

Paragraph 3.2.1. Performance Characteristics. This paragraph shall state what the item shall do in terms of complete functional characteristics only and shall specify upper and lower limits for each performance characteristic. These characteristics shall be expressed as requirements that must be achieved and not as goals or best efforts.

Paragraph 3.2.2. Physical Characteristics. This paragraph shall include the following, as applicable:

a. Weight limits of the item.

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b. Dimensional and cube limitations, operator station layout, ingress, egress, and access for maintenance.

c. Durability factors to indicate degree of ruggedness.

d. Health and safety criteria, including consideration of adverse explosive, mechanical, and biological effects. Included in this criteria are the toxicological effects of the equipment on the operation and the adverse effects of any electromagnetic radiation that might emanate therefrom.

e. Vulnerability factors including consideration of chemical and radiological operations, electromagnetic radiation, and fire.

Paragraph 3.2.3. Reliability. Reliability shall be stated in quantitative terms such as mean time between failure (MTBF), defining the conditions under which the requirements are to be met. This paragraph may include a reliability apportionment model to support apportionment of reliability values assigned to major components for their share in achieving desired item reliability.

Paragraph 3.2.4. Maintainability. This paragraph shall specify the quantitative maintainability requirements. The requirements shall apply to maintenance in the planned maintenance and support environment and shall be stated in quantitative terms. Examples are:

a. Time (e.g., mean and maximum downtime, mean and maximum time to repair, mean time between maintenance actions).

b. Rate (e.g., maintenance manhours per specific maintenance action, maintenance hours per operating hours, frequency of preventive maintenance).

c. Maintenance complexity (e.g., number of people and skill levels, variety of support equipment).

d. Maintenance costs (e.g., maintenance costs per operation hours, man-hours per overhaul).

Paragraph 3.2.5. Availability. The proportion of the time that equipment is up and running. It is expressed quantitatively as the up time over the up time plus the down time, or MTBF over MTBF plus MTTR.

Paragraph 3.2.6. Environmental Conditions. This paragraph shall include both induced and natural environmental conditions expected to be encountered by this item during storage, shipment, and operation. It shall include factors such as shock, vibration, and temperature.

Paragraph 3.2.7. Transportability. This paragraph shall include requirements for transportability of the equipment to the facility and within the facility. Any abnormal requirements for transportability will be identified here.

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Paragraph 3.3. Design and Construction. This paragraph shall specify minimum item design and construction standards which have general applicability and are applicable to major classes of equipment or are applicable to particular design standards. To the maximum extent possible, these requirements shall be specified by reference to the established military standards and specifications. In addition, this paragraph shall specify criteria for the selection and imposition of federal, military, and contractor specifications and standards.

Paragraph 3.3.1. Materials, Process, and Parts. This paragraph shall specify those item-peculiar requirements governing use of materials, parts, and processes to be utilized in the design of the item. It shall also contain specifications as necessary for particular materials and processes to be utilized in the design of the item. Special attention shall be directed to prevent unnecessary use of strategic and critical materials. In addition, requirements for the use of standard and commercial parts listed in the qualified products list shall be specified in this paragraph.

Paragraph 3.3.2. Tooling. This paragraph shall specify requirements for special tooling and shall be related as required to paragraph 3.2.3, 3.2.4, and 3.2.5.

Paragraph 3.3.3. Identification and Marking. This paragraph shall cover the requirements for the use of color function and identification coding of electrical, hydraulic water lines, etc. This paragraph shall also cover the requirements for nameplates for identification and for operation and safety. Requirements for serialization and nomenclature shall not be included in this paragraph.

Paragraph 3.3.4. Interchangeability. This paragraph shall specify the requirements for those components of the item to be interchangeable. Entries in this paragraph are for the purpose of establishing a condition of design and do not define the conditions of interchangeability that are required by the assignment of a part number.

Paragraph 3.3.5. Safety. This paragraph shall specify requirements to preclude or limit hazards to personnel and other equipment. To the extent practicable, these requirements shall be imposed by citing established and recognized standards. Appropriate paragraphs of the functional criteria specification shall be cited. Limiting safety characteristics peculiar to the item due to hazards in assembly, disassembly, test, transport, storage operation, or maintenance shall be stated when covered neither by standard industrial or service practices nor by the functional criteria specification. "Fail-safe" and emergency operating restructions shall be included where applicable. These shall include interlocks and emergency and standby circuits required to prevent injury.

Paragraph 3.3.6. Human Performance/Human Engineering. Human engineering requirements for the item should be specified herein and applicable documents included by reference. This paragraph should also specify any special or unique requirements, e.g., constraints on allocation of functions to personnel and personnel/equipment interactions. Included should be those specific areas, stations, or equipment which would require concentrated human engineering attention due to the sensitivity of the operation or criticality of the tasks, i.e., those areas where the effects of human error would be particularly serious.

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Paragraph 3.3.7. Standards of Manufacture. This paragraph, when applicable, shall include those standards that because of their significance, must be set forth as a requirement for the manufacture of the item. Requirements specified herein shall be to the level of detail necessary to clearly establish limits for the inspections included in Section 4.

Paragraph 3.3.8. Facilities. This paragraph shall address the requirements for housing the equipment including clear span and column-free areas, floor level requirements, floor load, clear ceiling heights, etc.

Paragraph 3.3.9. Utility Consumption. This paragraph shall delineate the utility consumption for the item for each operational rate. These requirements shall be listed and include: electrical power, fuel, oil, gas, water steam, etc.

Paragraph 3.4. Major Component Characteristics. This paragraph shall include a subparagraph for each major component listed in 3.1.3. The paragraph shall state the performance and physical characteristics for each major component. In stating requirements for the various components, it should be recognized that verification may necessarily need to be accomplished following the delivery, installation, and checkout of the end item.

Section 4. Quality Assurance Provisions. Requirements for formal tests/verifications of the item performance and physical characteristics shall be specified in this section.

Paragraph 4.1. General. This paragraph shall, as applicable, provide general information pertinent to tests and inspections not covered elsewhere in Section 4, such as location or conditions for qualification testing, requirements for special testing of item components, etc.

Paragraph 4.1.1. Special Tests and Examinations. This paragraph shall cover the testing routine, sequence of tests, data required, etc., for all testing requirements other than acceptance inspection.

Paragraph 4.2. Quality Conformance. This section shall list all examinations and tests required to verify the requirement of Sections 3 and 5 have been achieved in the item. These examinations and tests shall include or reference, as appropriate.

a. Tests and checks of the performance and reliability requirements.

b. Measurement or comparison of specified physical characteristics.

c. Verification, with requirements for warranties,

d. Method of confirming that the item, as fabricated and assembled, complies with the requirements of the equipment technical specification and the drawings.

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Section 5. Preparation for Delivery. This section shall include applicable requirements for preservation, packaging, and packing of the item and marking of packages and containers. The degree of assembly or assembly confiruation for shipment of equipment from the vendor plant shall be specified.

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Custodian:
Army - AR

Review activities:
Army - AL, CE

Preparing activity:
Army - AR

(Project CMAN-A002)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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